

The empirical recurrence for OEIS A258520, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A258520 records “Empirical recurrence of order 94” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 243$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 12$, so the recurrence is proved for every $n \geq 106$. Its last coefficient is $c_{94} = 2 \neq 0$, so no further tail satisfies anything shorter; any order-94 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A258520 is “Number of $(n+2) \times (2+2)$ 0..1 arrays with every 3×3 subblock sum of the two sums of the diagonal and antidiagonal minus the two sums of the central column and central row nondecreasing horizontally and vertically.”. It has offset 1 and begins

2371, 7334, 13616, 23785, 45771, 91023, 169230, 300889, ...

The entry states:

Empirical recurrence of order 94 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 15:26:06, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 243$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 243$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 94: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 12$ is the first whose minimal order is exactly the stated 94.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 486$ exact terms from $n = 12$ on, it returns order 94 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{94} c_i a(n-i),$$

c_1	4	c_{25}	−45606	c_{49}	377129	c_{73}	17939
c_2	−7	c_{26}	−5117	c_{50}	−35462	c_{74}	64112
c_3	8	c_{27}	18545	c_{51}	310101	c_{75}	−19266
c_4	0	c_{28}	−156006	c_{52}	1143602	c_{76}	−117062
c_5	11	c_{29}	70447	c_{53}	148486	c_{77}	−57715
c_6	−81	c_{30}	130459	c_{54}	−305042	c_{78}	−10809
c_7	153	c_{31}	−23405	c_{55}	139388	c_{79}	−20122
c_8	−197	c_{32}	70629	c_{56}	−620838	c_{80}	6907
c_9	26	c_{33}	324432	c_{57}	−965792	c_{81}	22756
c_{10}	27	c_{34}	−168344	c_{58}	56670	c_{82}	9805
c_{11}	695	c_{35}	−243187	c_{59}	166850	c_{83}	1919
c_{12}	−1422	c_{36}	74242	c_{60}	−83201	c_{84}	1147
c_{13}	2115	c_{37}	−335993	c_{61}	766546	c_{85}	−1859
c_{14}	−480	c_{38}	−497150	c_{62}	691043	c_{86}	−2010
c_{15}	−1207	c_{39}	289934	c_{63}	−84018	c_{87}	−763
c_{16}	−3182	c_{40}	284759	c_{64}	−7939	c_{88}	−128
c_{17}	7172	c_{41}	−83412	c_{65}	−8377	c_{89}	385
c_{18}	−13239	c_{42}	741980	c_{66}	−632739	c_{90}	170
c_{19}	4050	c_{43}	554370	c_{67}	−416265	c_{91}	−28
c_{20}	10048	c_{44}	−372951	c_{68}	17017	c_{92}	−22
c_{21}	7666	c_{45}	−167801	c_{69}	−78555	c_{93}	−4
c_{22}	−19813	c_{46}	−38173	c_{70}	34198	c_{94}	2
c_{23}	54369	c_{47}	−1076158	c_{71}	344037		
c_{24}	−20649	c_{48}	−415774	c_{72}	191021		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 106$, and it is the recurrence OEIS A258520 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 12$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{94} - c_1x^{94-1} - \dots - c_{94}$. Its constant term is $-c_{94} = -2$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 94 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 94, so $\deg q = 94$, and it is monic. It holds from some index t on. From $\max(t, 12)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 94, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 94, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 2.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-94 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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